Consumer360 – Customer Segmentation & RFM Analysis

Project Overview

Project Title: Customer Segmentation using RFM Analysis
Product Name: Consumer360
Domain: Data Analytics / Business Intelligence

Business Problem

A mid-sized e-commerce company was running generic marketing campaigns that were not effective. The business needed a data-driven way to:

- Identify **high-value customers (Champions)** for premium engagement
- Identify **churn-risk customers** for targeted retention
- Enable decision-makers to monitor customer behavior dynamically

Solution Summary

An end-to-end analytics solution was built using **Python and Power BI** to segment customers using **RFM (Recency, Frequency, Monetary) Analysis** and visualize insights in an interactive dashboard.

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2. Tools & Technologies Used

- **Python (Google Colab)** – Data loading, cleaning, RFM calculations
- **Pandas Library** – Data manipulation and aggregation
- **Power BI Desktop** – Dashboard creation and visualization
- **Excel / CSV** – Input and output data storage

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3. Dataset Description

The dataset represents customer transaction-level data from an e-commerce platform.

Key Columns Used

- **CustomerID** – Unique identifier for each customer
- **InvoiceDate / OrderDate** – Date of purchase
- **Monetary** – Total revenue generated by the customer
- **Frequency** – Number of purchases made by the customer

- **Recency** – Days since the customer's most recent purchase
- **Country** – Customer's country

Multiple sheets existed in the source file (raw, cleaned, dashboard-ready). For analysis, only the **cleaned transactional sheet** was used.

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4. Methodology

4.1 Data Preparation (Python)

1. Loaded the cleaned dataset into Google Colab using Pandas
2. Converted date columns to proper datetime format
3. Ensured no missing or invalid values in key analytical columns

4.2 RFM Metric Calculation

RFM analysis was performed as follows:

- **Recency (R):** Number of days since the customer's most recent purchase
- **Frequency (F):** Total number of transactions per customer
- **Monetary (M):** Total revenue contributed by each customer

Each metric was scored on a **1–5 scale** using quantile-based binning.

4.3 Customer Segmentation

Customers were grouped into meaningful segments based on RFM scores, such as:

- **Champions** – High value, recent, frequent customers
- **Loyal Customers** – Frequent buyers with strong engagement
- **At Risk** – Previously valuable customers showing inactivity
- **Churn Risk** – Low recency and frequency customers

The final RFM output was exported and used as the input for Power BI.

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5. Dashboard Development (Power BI)

5.1 Key Visuals Created

KPI Cards

- **Total Customers** – Count of unique customers

- **Average Revenue per Customer** – Average monetary value
- **Average Purchase Frequency** – Mean purchase frequency

Charts & Tables

- **Customer Distribution by RFM Segment (Bar Chart)**
  Shows how customers are distributed across segments
- **RFM Summary Table**
  Displays average Recency, Frequency, and Monetary values per segment
- **Country Slicer**
  Enables filtering the dashboard by country

5.2 Interactivity

All visuals respond dynamically to slicer selections, allowing regional or segment-specific analysis.

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6. Key Business Insights

- A small percentage of customers contribute a large share of revenue (Champions)
- Churn-risk customers can be identified early based on declining recency and frequency
- Regional filtering helps marketing teams design location-specific strategies

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7. Business Impact

- Enables **targeted marketing campaigns** instead of generic promotions
- Helps improve **customer retention** and **lifetime value (CLV)**
- Supports data-driven decision-making for sales and marketing teams

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8. Project Outcome

- Successfully built an **end-to-end customer segmentation solution**
- Delivered an **interactive Power BI dashboard** ready for business use
- Demonstrated practical skills in **Python, analytics, and BI visualization**

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9. Conclusion

The Consumer360 project demonstrates a complete analytics workflow—from raw data processing to actionable business insights—making it a strong, industry-relevant portfolio project for data analytics roles.

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10. Google Colab (Python) – Full Code & Explanation

This section explains **exactly what code was used in Google Colab**, why it was used, and what output it produced. This is written for **absolute beginners**.

10.1 Opening Google Colab

Go to https://colab.research.google.com

10.2 Import Required Libraries

import pandas as pd

import numpy as np

from datetime import datetime

Why this code is used:

- pandas → for data loading, cleaning, and analysis
- numpy → for numerical calculations
- datetime → for date-based calculations (Recency)

10.3 Upload Dataset to Colab

from google.colab import files

uploaded = files.upload()

Explanation:

- Opens file upload dialog
- You uploaded your **sales / transaction dataset (CSV or Excel)**

10.4 Load Dataset into Pandas

```
df = pd.read_csv('sales_data.csv')
# OR if Excel
df = pd.read_excel('sales_data.xlsx')
df.head()
df.info()
```

Explanation:

- Loads data into memory
- head() → preview first 5 rows
- info() → checks column names, data types, missing values

10.5 Data Cleaning Steps

Convert Date Column

```
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
```

✓ Required for Recency calculation

Remove Missing Customer IDs

```
df = df.dropna(subset=['CustomerID'])
```

✓ RFM analysis requires valid customers

Create Total Sales Column

```
df['TotalPrice'] = df['Quantity'] * df['UnitPrice']
```

✓ Monetary value calculation

10.6 RFM Calculation

Define Reference Date

```
snapshot_date = df['InvoiceDate'].max() + pd.Timedelta(days=1)
```

✓ Latest transaction date + 1 day

Calculate RFM Metrics

```
rfm = df.groupby('CustomerID').agg({
    'InvoiceDate': lambda x: (snapshot_date - x.max()).days,
    'InvoiceNo': 'count',
    'TotalPrice': 'sum'
}).reset_index()
```

Rename Columns

```
rfm.columns = ['CustomerID', 'Recency', 'Frequency', 'Monetary']
```

10.7 RFM Scoring (1–5 Scale)

```
rfm['R_Score'] = pd.qcut(rfm['Recency'], 5, labels=[5,4,3,2,1])
rfm['F_Score'] = pd.qcut(rfm['Frequency'].rank(method='first'), 5, labels=[1,2,3,4,5])
rfm['M_Score'] = pd.qcut(rfm['Monetary'], 5, labels=[1,2,3,4,5])
```

Explanation:

- Higher **Recency score = more recent purchase**
- Higher **Frequency score = more purchases**
- Higher **Monetary score = higher spending**

10.8 Combine RFM Scores

```
rfm['RFM_Score'] = rfm['R_Score'].astype(str) + rfm['F_Score'].astype(str) +
rfm['M_Score'].astype(str)
```

✓ Creates combined customer score

10.9 Customer Segmentation Logic

```
def segment_customer(row):
    if row['R_Score'] == '5' and row['F_Score'] == '5':
        return 'Champions'
    elif row['R_Score'] >= '4' and row['F_Score'] >= '4':
```

```
        return 'Loyal Customers'
    elif row['R_Score'] <= '2' and row['F_Score'] <= '2':
        return 'At Risk'
    else:
        return 'Others'


rfm['Segment'] = rfm.apply(segment_customer, axis=1)
```

Business Meaning:

- **Champions** → High-value customers
- **Loyal Customers** → Repeat buyers
- **At Risk** → Possible churn

10.10 Export RFM Output for Power BI

```
rfm.to_csv('rfm_output.csv', index=False)
files.download('rfm_output.csv')
```

✓ This file was uploaded into **Power BI**

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11. How This Pipeline Works End-to-End

Raw Sales Data

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Data Cleaning (Python)

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RFM Calculation

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Customer Segmentation

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RFM Output CSV

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12. Why This Project Is Industry-Ready

✓ Follows real-world analytics pipeline
✓ Uses Python for logic
✓ Uses Power BI for visualization
✓ Beginner-friendly yet production-aligned